

Derbyshire Shared Care Pathology Guidelines

Diagnosis of Diabetes Mellitus in Adults

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Type 1 Diabetes Mellitus in Adult

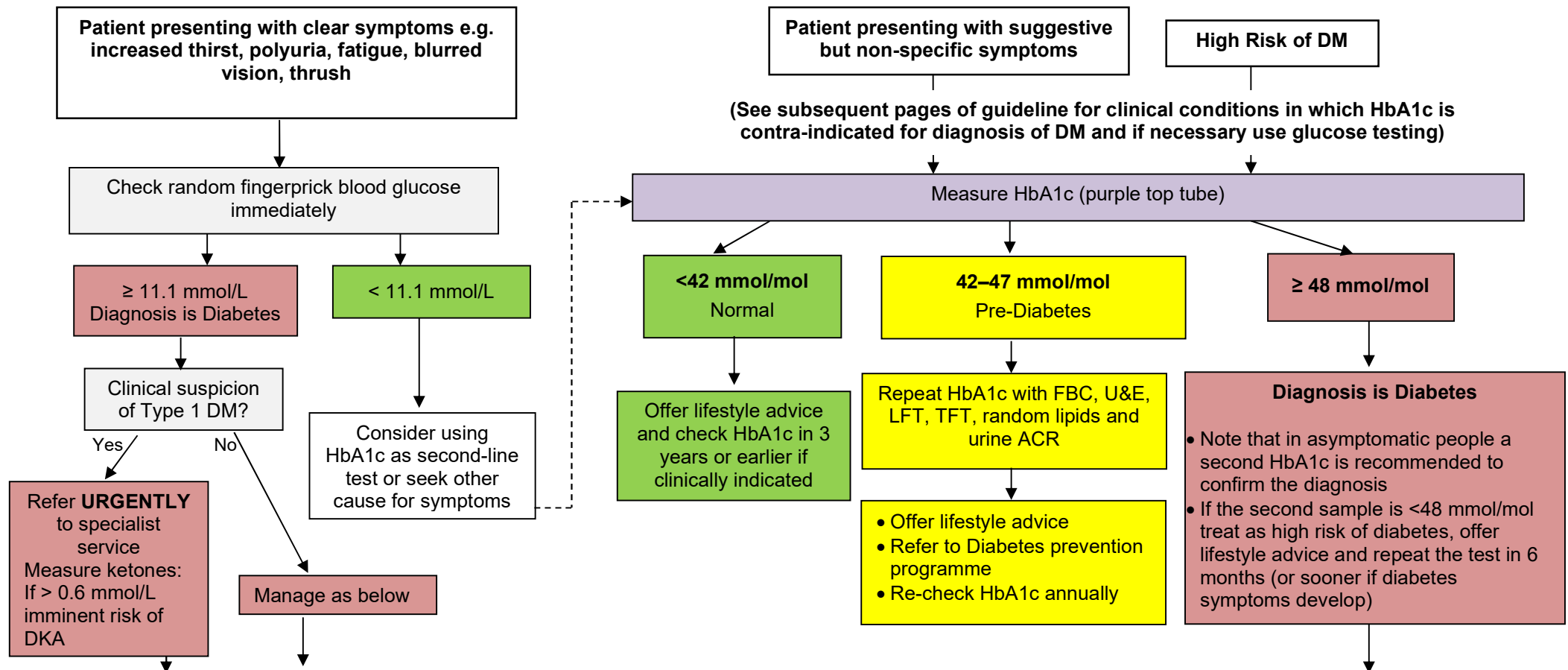
Age or BMI alone should not be used to exclude or diagnose Type 1 Diabetes in adults.

1. Make an initial diagnosis of type 1 diabetes on clinical grounds in adults presenting with hyperglycaemia. Bear in mind that people with type 1 diabetes typically (but not always) have 1 or more of:
 - Rapid weight loss
 - BMI below 25kg/m²
 - Personal / family history of auto-immune disease.
 - Ketones > 0.6 mmol/L

Note there are an increasing number of young people <40years being diagnosed with T2DM. Young people with T2DM typically have a strong family history of diabetes, are more likely to be obese or be from a minority ethnic background. If there is any doubt about the type of diabetes a person has please refer in for specialist review, diabetes autoantibodies and assessment. Missing a diagnosis of T1DM can have fatal consequences.

2. **Refer urgently** into diabetes services for assessment and management if random fingerprick glucose is ≥ 11.1 mmol/L (or if patient presents fasting, glucose ≥ 7 mmol/L) and any suspicion of Type 1 diabetes. If ketones > 0.6 mmol/L there is imminent risk of DKA.
3. If someone with an established diagnosis of Type 2 diabetes (>3 years duration) is suspected of having Type 1 diabetes, a C-peptide and glucose can be checked. C-peptide <600 pmol/l raises the possibility of Type 1 diabetes. Please refer into diabetes services.

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What to do with patients diagnosed with Diabetes:

- Type 1: Refer to specialist service immediately for insulin and education. Note ketones can be negative in T1.
- Type 2: Check FBC, U&E, LFT, TFT and lipids and ACR if not already done

Offer contraception advice to all women of childbearing age and refer to pre-conception service if considering pregnancy

Offer explanation, support and either printed Diabetes UK information booklet or signpost to their learning zone ([Learning Zone - Discover more about your diabetes.](#) | [Learning Zone](#))
 Consider referral to Diabetes Education course (e.g. Diabetes and You) or NHS Pathway to remission [NHS England » NHS Type 2 Diabetes Path to Remission Programme](#)
 Check feet and blood pressure
 Refer to retinopathy screening
 Discuss and agree individualised treatment targets

Urgent referrals

Any Suspected new Type 1 DM needs assessment by secondary care the SAME day.

Using HbA1c to diagnose Type 2 Diabetes

In 2011, the WHO recommended the use of HbA1c as a test to diagnose (rather than just monitor) diabetes, using a threshold of 48 mmol/mol. This has subsequently been endorsed by Diabetes UK. However, the recommendation comes with a number of caveats, which make its incorporation into routine practice far from straightforward.

HbA1c and glucose testing are not entirely concordant, and individual patients may fall into different categories with different tests.

It is therefore recommended to use only one test, wherever possible, to avoid confusion.

- Diagnosis should NOT be made on the basis of a point-of-care (fingerprick) HbA1c, a laboratory result is required.
 - RDH uses a HPLC method. Note that homozygous Hb variants (HbS, HbC and HbD) are also detected by this method and will be confirmed by an alternative method if found.
 - CRH uses a Total Affinity method. Hb variants are not detected by this method
 - QHB uses a HPLC method. Hb variants are detected by this method and if present will be indicated on the report.

Selecting the right patient group

The primary value of HbA1c is in people at high risk of type 2 diabetes, and/or suggestive but non-specific symptoms, where a fasting glucose or GTT might otherwise be considered.

Patients with clear symptoms of diabetes should have a random glucose measured.

HbA1c is a lagging indicator and may be unreliably low in patients with acute onset (usually type 1) diabetes. Its use for diagnosis is therefore **contraindicated in the following patient groups:**

- Children and young people
- Clinical picture of type 1 diabetes (any age): low BMI, weight loss, ketones
- Symptoms for <2 months
- Pregnancy
- Recent use of drugs that cause hyperglycaemia, e.g. steroids
- Recent acute pancreatitis or pancreatic surgery

Co-morbidities and treatments invalidating HbA1c

HbA1c is determined by the average level of glycaemia and by the length of time haemoglobin in red blood cells is exposed to glucose. Conditions that significantly affect the average age of circulating red cells will invalidate HbA1c. This applies as much to its

use in monitoring as in diagnosis, although it has generally been incorrectly ignored in clinical practice for many years.

HbA1c is **contraindicated in the following situations (which may be subclinical and unrecognised)**:

- Haemolytic anaemia (HbA1c is exquisitely sensitive to even mild haemolytic disorders)
- Iron deficiency anaemia
- Vitamin B12 and/or folate deficiency
- Recent blood donation, severe blood loss or transfusion
- Venesection
- Hereditary spherocytosis
- Haemoglobinopathies: see below
- Drugs: Dapsone, erythropoietin, anti-HIV, anti-hepatitis C, high dose vitamin C or E
- Chronic kidney disease (CKD4 or 5)
- Liver disease
- Alcoholism
- Severe hyperlipidaemia
- Chronic malaria

Haemoglobinopathies

Haemoglobin variants have variable effects on HbA1c results. Most common variants do not directly affect the measurement of HbA1c itself. It is more common that a variant shortens red cell survival time. Some rare variants (which may occur in any ethnic group) may directly interfere in HbA1c measurement, usually giving falsely low results. Disorders causing high levels of fetal Hb (HbF), such as thalassaemias, give falsely low results.

Interpreting HbA1c results

Since most confounding factors tend to lower HbA1c, and those that raise it do so by a relatively small amount, a high HbA1c is a good indicator of diabetes (i.e. it has high specificity).

The higher the HbA1c the more likely diabetes is present.

HbA1c below 48 mmol/mol does not exclude diabetes, even if all of the above conditions are followed, due to the possibility of undiagnosed confounders. Where there is any doubt about the reliability of HbA1c it should not be used, and glucose testing following existing WHO criteria should be performed instead.

HbA1c reference ranges

Derbyshire pathology use 2 separate reference ranges for HbA1c tests, one for diagnosis and one for monitoring.

- **For monitoring purposes:**

NICE NG28 (Management of type 2 Diabetes in adults, 2020) states:

- Aim for 48 mmol/mol in most groups with diabetes

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- Adults with T2DM on insulin or drugs likely to cause hypoglycaemia, aim for 53 mmol/mol
- Targets should be individualised according to risk of complications vs risk of hypoglycaemia
- **For diagnostic purposes:**
 - If clear symptoms of diabetes, measure random glucose
 - HbA1c may be unreliably low in patients with acute onset (usually type 1) DM
 - Do not use for diagnosis in: children and young people; likely type 1 DM (any age); low BMI, weight loss, ketones; symptoms for <2 months; pregnancy; recent use of hyperglycaemic drugs, e.g. steroids; recent acute pancreatitis or pancreatic surgery
 - HbA1c is unreliable in any anaemia, recent blood donation, severe blood loss, transfusion, venesection, CKD4 or 5, liver disease, alcoholism
 - HbA1c between 42 - 47 mmol/mol are consistent with non-diabetic hyperglycaemia (NDH), unless the patient has known diabetes

References

1. Type 1 Diabetes in adults: Diagnosis and management. NICE NG17. [Overview | Type 1 diabetes in adults: diagnosis and management | Guidance | NICE](#)
2. The Management of Type 1 Diabetes in Adults. A Consensus Report by the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD). <https://diabetesjournals.org/care/article/44/11/2589/138492/The-Management-of-Type-1-Diabetes-in-Adults>
3. Management of type 2 Diabetes in adults. NICE NG28. [Overview | Type 2 diabetes in adults: management | Guidance | NICE](#)

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